Engineering Mechanics 1 · Unit_03 · UNIT 3

DRAFT — UE26ME142A · Unit_03 · UNIT 3 · CLASS 31 Beams Numerical

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

CLASS 31 Beams Numerical

1 CLASS 31 Beams Numerical

Study focus: CLASS 31 Beams_Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000102 · EV0000103

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Question 5/105.

Key details

- Find the reactions at A due to the uniform loading and the applied couple.
- Look at this beam.
- The name of the beam is.
- Cantilever beam because one end is fixed and the other end is free.
- Hence it is called as cantilever beam.
- The name of the support here is fixed support.
- It will have three reactions, two forces, one along X, one more along Y, and one more will be the resisting moment.
- That can be either taken as clockwise or it can be taken as clockwise or.
- They have asked us to find the reaction at A, nothing but Ax, A by then resting moment at A we have to calculate.
- if you look at this, this is UDL, uniformly distributed loading.
- When we talk about the free body diagram, we need to convert this UDL into a point load.
- How to do that?
- We know the formula W into L.
- W will be equal to 2 kN and L will be.
- Where exactly that will be acting?
- Midpoint is nothing but three by.
- Two, that will be one point five meters.
- And this is moment of a couple it's a free vector.
- For discussion you need to start with the feedbody diagram.

- Before that, there is a.
- We know the formula to find the resultant R will be equals to W into L.
- What is W density of the loading that is 2 and L is span over which it is acting three meters.
- Where exactly the six will be acting?
- Since it's a UBN, it will be acting at a limit point.
- and only one point five meter from here.
- Here it's a fixed connection.
- Two reactions one three reactions to our forces A then AY There will be one more resting moment M A.
- Next, six we are done with this, then there's no loading and here twelve kilonewton meter.
- That moment is acting.
- Start with the conditions of equilibrium.
- So the first one is sigma above.
- is equals to zero.
- If you have this sigma if x is equal to zero, there is only one hard certain four.
- That is e x.
- Next apply the second condition sigma f of.
- Y is equals to zero.
- Since a is upward, so plus a y, then six is downward minus six.
- This is equal to zero.
- Then a y is equal to six kilon.
- The third condition is sum of the moment of all the forces as well as the couple above the point is zero taking counterquest positive.
- Then moment of these two forces is zero, but MA is nothing but the resisting moment because of the support.
- That is different sum of the [ASR low confidence; check audio].
- Since I have considered here counterclockwise, so plus.
- Yum vector don't multiply the distance.
- Next is six kilonute, update the right hand thumb rule.
- It is thumb is going inward clockwise.
- One point five.
- Then next if you go like this, lastly we have twelve kilometer meter.
- Since it is in a counterclockwise direction, it should be taken as plus twelve.
- It's already forced into distance, so don't multiply the distance.
- It's really vector.
- This should be bigger than zero.
- Three kilonewton meter.
- What this minus sign indicates?
- We assume the existing moment here at A is in the counterclockwise direction, but actually that is in the clockwise direction.
- Three kilonewton meter clockwise.

Example or application from the lecture

- First, let's start with the beam.

- Before that let's discuss about the type of loading.
- When we consider sigma f and sigma f y we should not consider this because this is moment of a couple.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 05:00–09:59

These are the reactions at this about E.

Key details

- This is the next question 5/116.
- For the beam and the loading shown, determine the magnitude of force F for which the vertical reaction at A and B are equal.
- With this value of F, compute the magnitude of the pin reaction at A.
- Nothing but Ax will be equal to R B or one more way R B will be equal to Ax.
- Ah F is also a unknown.
- After that there's one more unknown that is Ax.
- And in this particular question, Ax will not be equal to 0 because this is the entire beam.
- Horizontal and vertical.
- This is a single beam.
- R B as well as F will be unknowns.
- So four unknowns means statically indeterminate.
- Yo what is vertical reaction at D is equals to reaction at the end.
- For what value of this F, if A is equal to R B we have to calculate.
- Just project all these points vertically downward and start with the B.
- we have this is E and this is B.
- Only the points load in the free body line, so I'm going to call it as one.
- If this is what I need to find the result end of that loading, I'll call it as R1.
- R1 is equal to since it is in the form of triangle, it will be half into base into height.
- W by two.
- Half into what is W intensity of 2 into length.
- Span over exactly UEL acting is 2.
- Next way exactly it will be acting from here to L by three from here.
- R1 that is equal to 2 kilonewt at a distance of from here 2 by 3.
- From here 2 times of 2 by 3.
- Next, here we have two repezoidal loading.
- One wall span of four meters and one more one more wall span of three meters.
- Then how to divide?
- Just draw a horizontal line here.
- R2 stands for the resultant point load of the rectangle.
- Formula will be W into L.
- What is W intensity?
- It will be eight kilonewton.

- Where exactly this way?
- 8 kilonewton will be at the midpoint.
- Since it is four meter, 4 by 2 will be equal to 2 meters, that will be less than the point A.
- Span will be 2 meters.
- This is two meters from here to here.
- Next, R three stands for the result of the triangle.
- Half into base.
- That so that will be four span into height will be equal to five.
- Total is five.
- That you'll get us three.
- Where exactly this tool twelve kilonewton will be here?
- From here two times the deaf L by three from here.
- L by three.
- How much will get?
- That will be left side of the point.
- T by point three.
- Two times the tough.
- From this side, two times the duff.
- Four by three.
- This is the distance.
- Three you will get it.
- Then from here eight by three.
- From here eight by three means to the calculation, you'll get it as two point double six seven.
- Two point double six seven means in between eight and.

Example or application from the lecture

- If you don't consider the given condition a y is equal to r b, this will become statically indeterminate because ax a y r.
- Remember we have to consider if I is equals to RB.
- Let's get into the solution part.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 10:00–14:59

Point E we'll get this slowly.

Key details

- This is twelve kilometer.
- Remember it is between eight and eight.
- The distances from here to here it will be 2.67.
- Next, A we have a pin correction, two reactions, AY, then one more will be horizontal.
- This will be A.

- I have taken it left side because anyhow here it is that to balance that A should be towards left.
- One meter we don't have any loading, next comes the trap side loading.
- Again draw descent line passing through the this will be the fourth one and this will be the fifth one.
- Descendant of fourth one fourth will be because since it is a rectangle, W into L will be the descendant.
- What is W three.
- Because this is the height of the rectangle and L is three.
- Where exactly this 9 will be acting?
- That we have to calculate midpoint.
- Nothing but from this point, 1.5 meter.
- Then from here to here, one point five meter we have this R4 of nine kilo Newton.
- Next, after this, we need to find the resultant R5 will be equals to.
- Triangle so half into base into height base is three and the height will be total five minus three is the height of the triangle.
- That will be two.
- Where exactly this three will be acting from here L by three from here to L by three.
- L by three means four.
- L by three.
- L A ye is three.
- Three by three will be equal to one meter.
- Nothing but it will be left side of this nine kilonewtons.
- This is three kilonewt and the distance will be one meter.
- Next we have to add this three meters.
- We don't have any loading for the next two meters, take that as it is.
- Next year it's roller connection so only by the reaction RB.
- Yes, take it as it is.
- We are done with the free body diagram.
- Remember we have to write AY is equal to RB.
- Next we have to apply the conditions of equilibrium.
- Start with the first condition sigma f of x is equals to zero.
- 2 kilonewt is right side, so plus 2, and this is left side, AX.
- AX will be equals to 2 kilonewt.
- Our assumption is correct.
- We assume AX is towards left.
- That is correct.
- Next is moment equation.
- I will for moment about point is zero.
- Taking counterclockwise is positive.
- Start with this two kilonut.
- Thump is coming outward because the force is acting in this direction, so curl your fingers in this direction.

- It is counterclockwise, so plus two into perpendicular distance will be this is the line of action of the force.
- And this will be the perpendicular distance that is vertical by three.
- We are then the moment of this next is this A kilo Newtons.
- It is acting in the counterclockwise direction, so plus yet.
- And what will be the distance?
- The distance from here to here is two meters.
- Three minus two will use one meter distance.
- Next is twelve kilon, that is counterclockwise.
- Plus twelve distances from year to year it is three.
- T minus two point double six seven.
- Three kilonewton is acting in a clockwise direction.
- Next nine kilon counterclockwise.
- Minus nine so nine kilometer clockwise.
- Then comes RB.
- RB is in a counterclockwise direction.

Example or application from the lecture

- You are calculating moment about E, so consider this part of your palm RTA, and you have to apply the right hand thumb rule.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 15:00–19:59

From here to here it will be two plus three five plus one six.

Key details

- Then F is in a clockwise direction.
- This will be 3 plus 25.
- 5 plus 3 8.
- 8 plus 1 9.
- This should be equal to 0.
- Simplify this, you'll get the value of you'll give one equation 6 RB minus 9 F is equal to 15.
- You'll get this expression condition of equilibrium that is sigma f of y is equal to zero.
- If you apply this.
- 8 is acting downward, so minus 8, then 12 is acting downward minus 1, 3 is acting downward minus 3, then 9 downward, minus 9, a y I left, so plus a y.
- Then plus RB.
- Minus F is equal to zero.
- Since E is equal to RB.
- Then you'll get two RB minus F is equal to 26.
- This is the equation two.
- Once you solve one and two, from one and two, you'll get the value of RPS.

- One eight point one kilo Newton.
- Then F as one zero three six.
- One zero point three six kilo Newton.
- This is the value of.
- AY that is one eighteen point one eight is with this I am going to conclude today's session.
- go through the assignment questions.
- If you come across any queries you can post your queries in PSU forum.

Example or application from the lecture

- Since it is horizontal, only to consider this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
CLASS 31 Beams_Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 31 Beams_Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 31 Beams_Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 31 Beams_Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_03/AV_Summaries/041_CLASS_31_Beams_Numerical__2ddd1ae1ec9c2c35.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 15 rendered pages in 033_Centroid__9f0e55503970de0e.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 033_UE26ME142A_Unit_3_Class_33_Lecture_PPT__ce948fc3cd6a59bf.slides.md have no readable `.slides.md` text; their original page images are included.
3. 3 rendered pages in 039_UE24CV141A__Unit_3_Beams__9fc34bbd5ca57988.slides.md have no readable `.slides.md` text; their original page images are included.
4. 3 rendered pages in 039_UE25CV141A_Unit_3_Class_48_Lecture_PPT__4d28f283532e284e.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 042_UE25CV141A_Unit_3_Class_53_Lecture_PPT__c20af5d61d14aaf0.slides.md have no readable `.slides.md` text; their original page images are included.
6. 5 rendered pages in 043_UE25CV141A_Unit_3_Class_54_Lecture_PPT__c7db1f404daea47d.slides.md have no readable `.slides.md` text; their original page images are included.
7. 6 rendered pages in 044_UE25CV141A_Unit_3_Class_55_Lecture_PPT__741ad989a632bc04.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 044_UE26ME142A_Unit_3_Class_44.1_Lecture_PPT__068ee897e4f8117e.slides.md have no readable `.slides.md` text; their original page images are included.
9. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
10. 6 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
11. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.

12. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
13. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
14. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
15. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare...
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0059. Sidecar variants compare...

C Revision and retrieval practice